

Wacky steps (30.07)

Today we prove the main thm: $G_K \cong G_{\mathbb{Q}_p} \Leftrightarrow K \cong \mathbb{Q}_p$

Prop 5.1: $K \cong \mathbb{Q}_p \Rightarrow G_K \cong G_{\mathbb{Q}_p}$

Proof: $G_{\mathbb{Q}_p}$ is small (cf my talk) & finite extensions are def.

Rest of the talk: $\Leftarrow$ First, how to prove that $K \cong \mathbb{Q}_p$?

Prop 4.11: $(K, v) \cong (\mathbb{Q}_p, v_p)$ iff

- (i) (K, v) is a hvf with $\text{char}(K) = 0$, $\text{char}(Kv) = p$;
- (ii) $v(p)$ is a minpos of vK (ie p is a unif)
- (iii) $vK \cong \mathbb{Z}$
- (iv) $Kv \cong \mathbb{F}_p$

Proof: (i) - (iv) are first-order & $(\mathbb{Q}_p, v_p) \models$ (i) - (iv), thus,
 $(K, v) \cong (\mathbb{Q}_p, v_p) \Rightarrow (K, v) \models$ (i) - (iv).

For $\Leftarrow$, we recall (or assume):

- Fact 5.2: an OAG $(\Gamma, +, <)$ is $\cong (\mathbb{Z}, +, <)$ iff
 $\exists \Delta \subseteq \Gamma$ s.t. Γ_Δ is div & $\Delta \cong \mathbb{Z}$ iff
 Γ has a minpos & $[\Gamma : u\Gamma] = u \ \forall u$.
- Fact (Sene?) if (F, w) complete, $\text{char}(F) = 0$, $wF \cong \mathbb{Z}$
& $w(p)$ minpos & $Fw = \mathbb{F}_p$, then
 $(F, w) \cong (\mathbb{Q}_p, v_p)$.
- AK/E (Thm 4.1): if (F, w) & (F', w') are hvf with
 $\text{char}(Fw) = 0$, then:
 $(F, w) \cong (F', w') \Leftrightarrow \begin{cases} wF \cong w'F \\ Fw \cong F'w' \end{cases}$

Now we do saturation - decomposition:

$(K, v) \& (K', v') \models (i) - (iv) \quad \mathbb{V}_I \text{ saturated.}$

$$K \longrightarrow K_{v_0} \xrightarrow[\substack{\langle p \rangle \cong \mathbb{Z} \\ p \text{ min pos}}]{(ii)} \mathbb{F}_p \leftarrow (iv)$$

$$K' \longrightarrow K'_{v'_0} \longrightarrow \mathbb{F}_p$$

$(K_{v_0}, \bar{v}) \cong (\mathbb{Q}_p, v_p)$ since sat $\Rightarrow$ complete.

also $(K'_{v'_0}, \bar{v}') \cong (\mathbb{Q}_p, v_p)$

$\Rightarrow (K', v'_0) \equiv (K, v_0)$ by AK/E

in $\mathbb{Q}_p$, we have $\mathcal{O}_{v_p} = \mathbb{Z}_p = \{x \mid \exists y \mid +px^q = y^q\}$ by J. Robinson

$\Rightarrow \bar{v}$ is def in K_{v_0} & $\bar{v}'$ is def in $K'_{v'_0}$, &

$(K', v'_0) \equiv (K, v_0) \Rightarrow (K'_{v'_0}, \bar{v}') \equiv (K_{v_0}, \bar{v})$

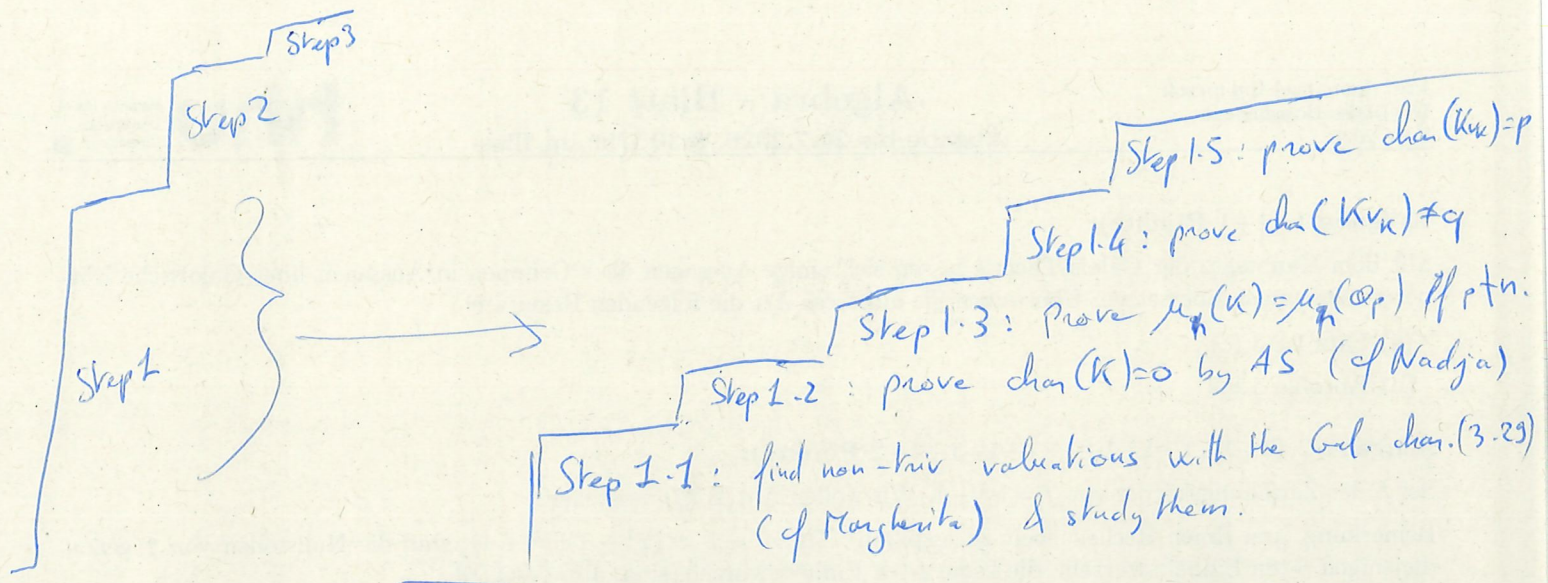
$\Rightarrow (K, v) \equiv (K', v')$.

Now we take $G_K \cong G_{\mathbb{Q}_p}$, we aim to find v on K henselian
s.t. $(K, v) \models (i) - (iv)$; in fact, we take v_K (cf Mordell)

Step 1: prove $\text{char}(K)=0$ & $\text{char}(K_{v_K})=p$ (in particular, non-trivial)

Step 2: standard: $K \rightarrow K_{v_0} \rightarrow K_{v_p} \rightarrow K_{v_K}$. Prove
that $(K_{v_0}, \bar{v}_p)$ has val group $\mathbb{Z}$ & residue finite.

Step 3: Wrap it up.



Lemma: if (F, v) is a hvf with $\text{char}(F_v) \neq q$, if M/F is Galois &
 $\text{Gal}(M/F) \cong \mathbb{Z}_q \rtimes (\mathbb{Z}_q \times C)$ with C finite of order prime to q ,
 then $R(M/F_v) = \{1\}$ & $I(M/F_v) \cong \mathbb{Z}_q \times \{1\}$.

Proof: $G = \text{Gal}(M/F) \cong \mathbb{Z}_q \rtimes (\mathbb{Z}_q \times C)$

$\Downarrow$

Δ

$I \cong q^s \mathbb{Z}_q \times \{1\}$ or $q^{s_1} \mathbb{Z}_q \rtimes (q^{s_2} \mathbb{Z}_q \times C')$ (no other closed non-trivial subgroups)

Δ

$R \cong \{1\}$ or $\{1\}$ (since R is prime to q)

Now $I \cong I/R \cong \text{Hom}(M/F_v, \mu_{M_v}) \leftarrow$ abelian

but $q^{s_1} \rtimes (q^{s_2} \mathbb{Z}_q \times C')$ is not abelian

$\Rightarrow I \cong q^s \mathbb{Z}_q \times \{1\}$.

Step 1.1: $q \neq p$. Take $G_q \leq G_{\text{sep}}$ a q -Sylow, we know

$G_q \cong \mathbb{Z}_q \rtimes \mathbb{Z}_q$, so $G_q \leq G_K$ also exists. Let $K_q = \text{Fix}(G_q)$

G_K has a q -Sylow of the form $\mathbb{Z}_q \rtimes \mathbb{Z}_q \Rightarrow F_{v_q}$ hensel tamely badly at q . (3.29)

So $\text{char}(K_q) \neq q$ & $v_q K$ is not q -div. Let $r_q = \dim_{\mathbb{F}_q}(v_q K / v_q K) \geq 1$.

w_q is the unique extension of v_q to K_q , & we have $\text{char}(K_q w_q) \neq q$ & $\dim_{\mathbb{F}_q}(w_q K_q / q w_q K_q) = r_q$ (since K_q/K is prime to q).

Lemma applied to $K_q^{\text{sep}} / K_q, w_q \Rightarrow R = \{1\}$ & $I = q^s \mathbb{Z}_q$.

$$\text{But } I \cong I/R \cong \text{Hom}(\text{div}(w_q K_q) / w_q K_q, \mu_q) \cong \mathbb{Z}_q^{r_q}$$

$$\text{So } \mathbb{Z}_q^{r_q} \cong q^s \mathbb{Z}_q, \text{ that is, } r_q = 1.$$

$$\text{We also have } \mathbb{D}/I \cong G/I \cong \text{Gal}(K_q w_q^{\text{sep}} / K_q w_q) \neq \{1\},$$

so $K_q w_q \not\equiv \text{SCF}$.

$$\text{More: } G_{K_q w_q} \cong C_{q^s} \rtimes \mathbb{Z}_q \triangleright C_{q^s}, \text{ so } F = \text{Fix}(C_{q^s})$$

$$\Rightarrow G_F \text{ is finite } \stackrel{\text{AS}}{\Rightarrow} G_F = \{1\} \text{ or } \#G_F = 2.$$

$$\text{if } G_F = \{1\}, \text{ then } s=0 \text{ & } G_{K_q w_q} \cong \mathbb{Z}_q.$$

$$\text{if } \#G_F = 2, \text{ then } q=2 \text{ & } s=1, \text{ & } F = (K_q w_q)^{\text{rc}}.$$

Let $h = K_q w_q$. We have:

$$\begin{array}{ccc} C_2 \rtimes \mathbb{Z}_2 & \begin{array}{c} h_1^{\text{alg}} C_2 \\ F = h^{\text{rc}} \\ \mid \\ \mathbb{Z}_2 \\ \mid \\ h \end{array} & \rightarrow \\ & \begin{array}{c} h^{\text{alg}} C_2 \\ F = h^{\text{rc}} \\ \mid \\ \mathbb{Z}_2 \\ \mid \\ h(\sqrt{x}) \\ \mid \\ C_2 \\ \mid \\ h \end{array} & \end{array}$$

$$\text{But } \sqrt{x} \mapsto -\sqrt{x} \in \text{Aut}(h(\sqrt{x})/h)$$

so $h(\sqrt{x})$ admits two orderings, $<_+$ & $<_-$.

$$\text{So } \begin{array}{c} h(\sqrt{x})_-^{\text{rc}} \quad h(\sqrt{x})_+^{\text{rc}} \\ \diagdown \quad \diagup \\ h(\sqrt{x}) \\ \mid \\ h \end{array}$$

But $G_K \cong C_2 \rtimes \mathbb{Z}_2$ has only one ~~solvable~~ normal subgroup of order 2, so $h(\sqrt{x})_+^{\text{occ}} = h(\sqrt{x})_-^{\text{occ}}$; & in this field $\sqrt{x} > 0$ & $\sqrt{x} < 0$, a contradiction.

thus $G_{K_q, w_q} \cong \mathbb{Z}_q$.

Step 1.2: $\text{char}(K_{q, w_q}) \neq q \Rightarrow \text{char}(K) \neq q$.

$\mathbb{Q}_p$ has a C_{p-1} extension $(\mathbb{Q}_p(\zeta_p)/\mathbb{Q}_p)$ which admits a C_p -ext not embeddable into a C_{p^2} -ext

$\Rightarrow G_{\mathbb{Q}_p}$ admits a solvable normal subgroup H with quotient $C_{p-1} \times C_p$
but no $H' \triangleleft H$ s.t. $G_{\mathbb{Q}_p}/H' \cong C_{p^2} \times C_{p-1}$.

$\Rightarrow G_K$ also

$\Rightarrow K$ admits a C_{p-1} -ext which itself admits a C_p -ext not embeddable into a C_{p^2} -ext

$\Rightarrow \text{char}(K) \neq p$.

(AS)

$\Rightarrow \underline{\text{char}(K) = 0}$

Step 1.3 we want to prove $\text{Gal}(K_q(q)/K_q) \cong \mathbb{Z}_q$
& use Lemma 3.4. (cf my talk)

Consider $\mathbb{Q}_{p, q} = \mathbb{Q}_p(\zeta_{q^\infty}, \sqrt[q]{p})$, we have

$$\begin{aligned} \text{Gal}(\mathbb{Q}_{p, q}/\mathbb{Q}_p) &\cong \text{Gal}(\mathbb{Q}_{p, q}/\mathbb{Q}_p(\zeta_{q^\infty})) \rtimes \text{Gal}(\mathbb{Q}_p(\zeta_{q^\infty})/\mathbb{Q}_p) \\ &\cong \mathbb{Z}_q \rtimes (\mathbb{Z}_q \times C) \end{aligned}$$

where $\#C \mid q-1$.

$$G_K \cong G_{\mathbb{A}_p} \Rightarrow \exists L_K \text{ with } \text{Gal}(L_K) \cong \text{Gal}(\mathbb{Q}_{p^{\infty}}/\mathbb{Q}_p)$$

Lemma applied to $L_K \Rightarrow R = \{1\}$ & $I \cong q^s \mathbb{Z}_q \times \{1\}$

$$\text{Now } G_{L_K/I} \cong D/I \cong \text{Gal}(L_{Kq}/K_{Kq}) \cong C_{q^s} \rtimes (\mathbb{Z}_q \times C)$$

So K_{Kq} admits a Galois ext with Gal grp $\mathbb{Z}_q$.

It is maximal pro- q since $G_{Kq/Kq} \cong \mathbb{Z}_q$.

So (K, v_q) fulfills the conditions of 3.4, & also $\mathbb{Q}_p, v_p$; so

$$y_{q^2} \in \mathbb{Q}_p \stackrel{3.4}{\Rightarrow} \mathbb{Q}_p \text{ admits a } C_{q^2} \times C_{q^2} \text{-ext}$$

$$\Updownarrow G_K \cong G_{\mathbb{A}_p}$$

$$y_{q^2} \in K \stackrel{3.4}{\Rightarrow} K \text{ admits a } C_{q^2} \times C_{q^2} \text{-ext}$$

$$\& \text{ we have } \{y_n \mid n \geq 1\} \stackrel{3.4}{\Rightarrow} \begin{cases} \mu_{p-1} & \text{if } p \neq 2 \\ \mu_2 & \text{if } p = 2 \end{cases}$$

Step 1.4 We can apply the methods above to obtain

$$y_{q^2} \in \mathbb{Q}_p(y_q) \Rightarrow \mathbb{Q}_p(y_q) \text{ admits a } C_{q^2} \times C_{q^2} \text{-ext}$$

$$\Updownarrow$$

$$y_{q^2} \in K(y_q) \Rightarrow K(y_q) \text{ admits a } C_{q^2} \times C_{q^2} \text{-ext}$$

Since $\mathbb{Q}_p(y_{q^{\infty}}) \neq \mathbb{Q}_p(y_q)$ (or y_4 if $q=2$), we

have $K(y_{q^{\infty}}) \neq K(y_q)$ & thus $K(y_{q^{\infty}})/K$ is infinite,

& $\text{Gal}(K(y_{q^{\infty}})/K) \cong \mathbb{Z}_q \times C$ for $\text{sc } C \text{ prim to } q$.

Assume $\text{char}(K_{v_K}) = q$, then $\text{char}(K_{v_q}) = 0$ since $\mathbb{Q}_{v_q} \supseteq \mathbb{Q}_{v_K}$.

So (K, v_q) is h.v.f. of equicharacteristic 0 $\Rightarrow \text{Gal}(K(\zeta_{q^n})/K) \cong \text{Gal}(K_v(\zeta_{q^n})/K_v)$

so $\text{Gal}(K_v(\zeta_{q^n})/K_v) \cong \mathbb{Z}_q \times C$ & $K_v(\zeta_{q^n})$ is q -closed

$$(\text{Gal}(K_v(\zeta_q)/K_v) \cong \mathbb{Z}_q)$$

$$\begin{array}{ccccc} K(\zeta_{q^\infty}) & \longrightarrow & K_v(\zeta_{q^\infty}) & \xrightarrow{\bar{w}} & K_v(K) \\ | & & | & \uparrow \text{add } q^{\text{th}} \text{ root of } 1 & \\ K & \longrightarrow & K_v & \xrightarrow{\bar{v}_K} & K_v(K) \end{array}$$

in a field of char q

in char q , $[F(\zeta_{q^{n-1}}) : F] \mid q^n$. Thus $K_v(K) \supset \zeta_{q^{n-1}} \forall n$.

We have $\gcd(\frac{q^{q^n}-1}{q^{q^{n-1}}-1}, q^{q^{n-1}}-1) = 1$; so there are infinitely many distinct primes q_n s.t. $q_n \mid q^{q^n}-1$; hence,

$\zeta_{q_n} \in K_v(K) \forall n$ & by lemma $\zeta_{q_n} \in K \forall n$; but

$\{\zeta_n \in K \mid \text{p.t.}\}$ is finite, a contradiction.

So, $\text{char}(K_v(K)) \neq q$.

Step 1.5 Apply Lemma to $K^{\text{sep}}/K_q, w$ unique extension of v_K to K_q

to get, as in step 1, $\dim_{\mathbb{Z}_q}(v_K(K)/v_K(K)) = 1$ (since $I \cong \mathbb{Z}_q \rtimes \mathbb{Z}_q \times \mathbb{Z}_q$)

& $K_v(K) \not\equiv \text{SCF}$ (since $\text{Gal}_{K_v(K)} \cong D/I \neq \{1\}$).

Apply Lemma again to $L/K, v_K$ as in step 1.3 to get $\text{Gal}(K_v(K)/K) \cong \mathbb{Z}_q$.

Now assume $\text{char}(K/K) = 0$ ~~the~~.

⇒

Step 1.5.4 contradiction

Step 1.5.3 $\exists!$ C_{p-1} -ext unramified at v_K

Step 1.5.2 $\exists!$ $C_{p-1} \times C_{p-1}$ -ext of K

Step 1.5.1 $v_K K$ is p -div

1.5.1: $G_p \leq G_K$ or p -Sylow, we know $C_p \cong \mathbb{Z}_p, \mathbb{Z}_p^{\times \mathbb{Z}_p}$
(lemma 3.15, of my talk). Now $R(\frac{K^{\text{sep}}}{K_p}, v_K) = \{1\}$
(residue char 0) & $I \cong I/R \cong \mathbb{Z}_p^{\times p} \triangleleft G_p$.
($r_p = \dim_{\mathbb{F}_p}(\frac{v_K K}{p v_K K})$).

If $r_p \geq 1$, then I is a non-trivial closed normal subgroup of G_p , & by Galois char of henselicity, K & $\mathbb{Q}_p$ both admit w.f. tamely body at p : impossible, so $r_p = 0$

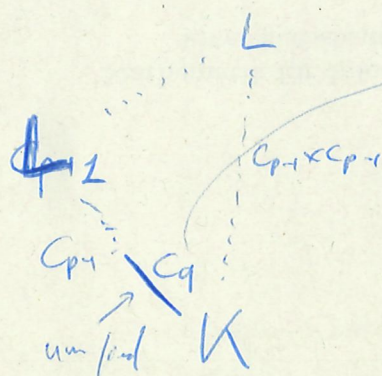
1.5.2 $\mathbb{Q}_p$ admits on $C_{p-1} \times C_{p-1}$ -ext: $\mathbb{Q}_p(\mathbb{F}_{p-1}^2, \mathbb{F}_p)$.

If $L, L'/K$ are both $C_{p-1} \times C_{p-1}$ -ext, then there is some

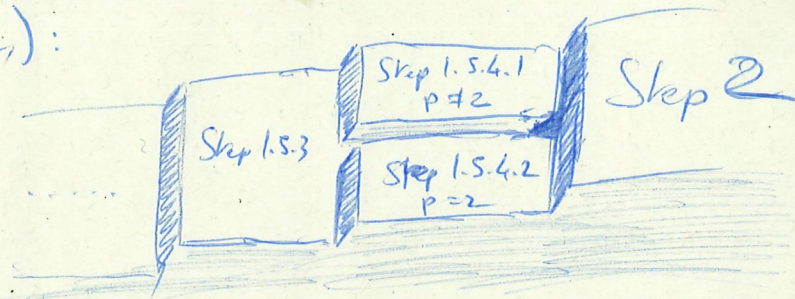
$\uparrow L/K$ with q -rank ≥ 3 where $q | p-1$; but

$G_q \cong \mathbb{Z}_q \times \mathbb{Z}_q$, so q -rank ≤ 2 .

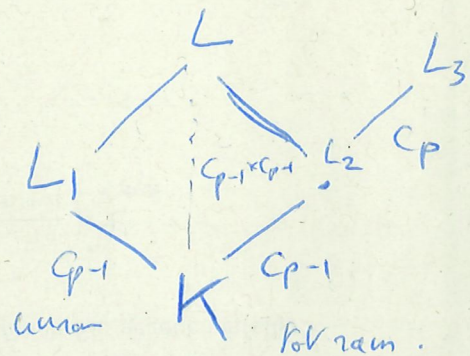
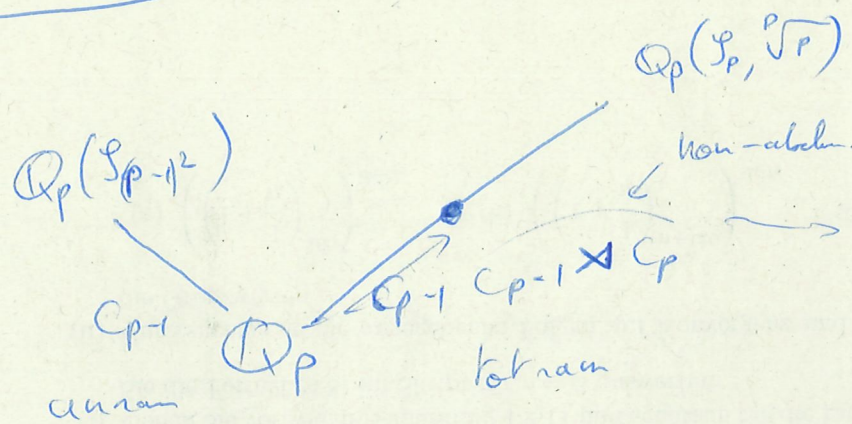
1.5.3 Because $\text{Gal}(K_K(q)/K_K) \cong \mathbb{Z}_q$:



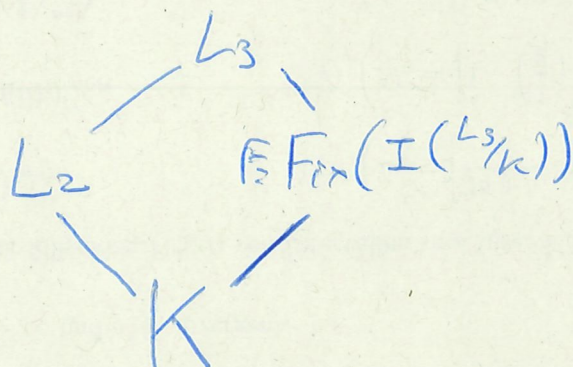
Step 1.5.4 (from above):
aside



1.5.4.1: $p \neq 2$



So $L_3/L_2 \cong C_p$; since $\sqrt[p]{k}$ is p -div, L_3/L_2 is unram.



So $L_3 = L_2 F$ & $\text{Gal}(L_3/k) = C_{p-1} \times C_p$ is abelian, a contradiction.

1.5.4.2: same idea, but slight differences, since $p=2$ you can't take $Q_2(\zeta_2)$ as an interesting extension, in fact, C_{p-1} will never be interesting. Fix: find $L_3/k \cong C_2^8 \rtimes (C_3 \times C_2)$. (we skip it).

Step 2: consider $K \rightarrow K_{v_0} \rightarrow K_{v_p} \rightarrow K_{v_K}$.
 Step 2.2: Pop's Lemma: $K_{v_0} \rightarrow K_{v_p}$ is discrete, residue field.
 Step 2.1: prove that $G_{K_{v_0}}$ is not the AGG of a field of char p ,
 & that $\overline{v_p} K_{v_0}$ is not p -div.

Step 2.1 is a mirror of Step 1.2: K has a C_p -ext M_1/K s.t. M_1 admits a C_p -ext M_2/M_1 , which doesn't embed into a C_{p^2} -ext. unique extension of v_0
 Push M_1 & M_2 to $K_{v_0} \subseteq M_{1,v_0'} \subseteq M_{2,v_0''}$; since $v_0 K$ is p -div (same arg as step 1.5), $M_{2,v_0''}/M_{1,v_0'} \cong C_p$ & doesn't embed into a C_{p^2} -ext (otherwise we could lift)
 $\Rightarrow G_{K_{v_0}}$ is not the AGG of a field of char p .

But: $(K_{v_0}, \overline{v_p})$ is a hnf of mixed char, $G_{K_{v_0}}$ is small,
 $\overline{v_p}(K_{v_0}) = \text{con}(v(p))$; if $\overline{v_p}(K_{v_0})$ is p -div, then

$\exists F$ of char p with $G_F \cong G_K$ (of Margalet); so $\overline{v_p}(K_{v_0})$ is not p -div.

Step 2.2 $G_{K_{v_0}}$ small $\Rightarrow (K_{v_0})^\times / (K_{v_0})^{\times p}$ finite $\xrightarrow{\text{Pop}} \overline{v_p}(K_{v_0})$ is either p -div $\times$ or discrete $\checkmark$, & K_{v_p} is finite.

(in particular, the standard is now $K \xrightarrow{\text{dis } 0} K_{v_0} \xrightarrow{\text{dis } 0} K_{v_p}$ (finite).)

Step 3:

Step 3.3 prove $v_K(p)$ is minimal

Step 3.2 prove that $v_K K \cong \mathbb{Z}$, i.e., $v_0 K$ is div.

Step 3.1: prove $K_{v_K} = \mathbb{F}_p$

3.1 is a minor of 1.3: $\{y_n \in K \mid p \nmid n\} = \begin{cases} \mu_{p-1} & \text{if } p \neq 2 \\ \mu_2 & \text{if } p = 2 \end{cases}$

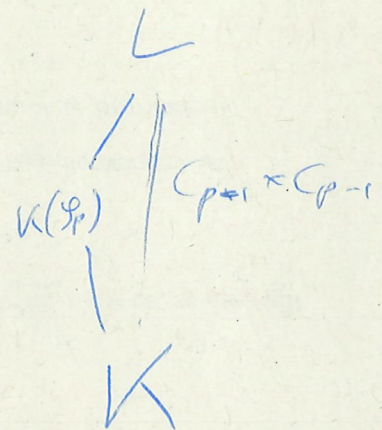
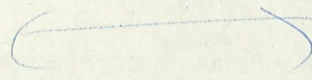
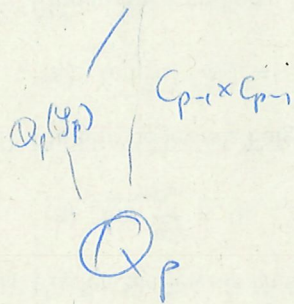
The only finite field of char p with these roots is $\mathbb{F}_p$.

3.2 we know that $v_K K$ is p -div & we know $(v_K K : q v_K K) = q$.

But $(\mathbb{Z} : q\mathbb{Z}) = q$, so what remains in $v_K K$ has to be of q -mult.

3.3 again consider:

$$F = \mathbb{Q}_p(\zeta_{p-1}, \zeta_p)$$



we have $f(L/K) = p-1$ & $e(L/K) = p-1$.

Say $v_K K \cong \mathbb{Z}$ has minpos 1, $v_K(p) \geq 1$, so

$v_L(p) = (p-1)v_K(p)$ (if $v_L L \cong \mathbb{Z}$ is non-lvl again at minpos 1).
 unique extn of v_K to L

degree Lemma (3.15, cf my talk): if $F/\mathbb{Q}_p$ finite dgh contains ζ_p , then:
 $\dim_{\mathbb{F}_p}(F^{\times}/F^{\times p}) = n+2$

$$\Rightarrow \dim(F^{\times}/F^{\times p}) = (p-1)^2 + 2.$$

general degree Lemma (3.16): if (L, v) is hvf with $\text{char}(L) = 0$, $\zeta_p \in L$, $v_L \cong \mathbb{Z}$ has minpos 1 & $v(p) < \infty$, & $L_v = \mathbb{F}_p$; then $\dim_{\mathbb{F}_p}(L^{\times}/L^{\times p}) = v(p) + 2$.

$$\Rightarrow \dim_{\mathbb{F}_p} (L^x / L^{x,p}) = \ell(L/K) v_L(p) + 2 = (p-1)(p-1) v_K(p) + 2$$

But by Kummer, "the unique extension of ~~degree~~ K with Gal group $C_{p-1} \times C_{p-1}$ has exactly $\mathbb{F}_p$ -dimension n' is a property of the Galois group; so, $\dim(F^x / F^{x,p}) = \dim(L^x / L^{x,p})$ & $v_K(p) = 1$.

Hence (K, v_K) is a huf with $\text{char}(K) = 0$ & $\text{char}(K v_K) = p$ (i), $v_K(p)$ is minpos (ii), $\forall K \in \mathbb{Z}$ (iii), & $K v_K \cong \mathbb{F}_p$ (iv)

(Bonus: proof of 3-16)

$$\begin{array}{c}
 K \xrightarrow{\text{divisible}} K v_0 \xrightarrow{\cong} K v \xrightarrow{\cong} K v_0 \\
 \downarrow \\
 K^x / K^{x,p} \cong K v_0^x / K v_0^{x,p} \cong \cancel{K v_0^x / K v_0^{x,p}} \cong F^x / F^{x,p}
 \end{array}$$

since $\text{cor}(v(p)) \in \mathbb{Z}$ is 1.
 coplan
 where $F/\mathbb{Q}_p$ is finite with $e(F/\mathbb{Q}_p) = \dim(\mathbb{F}_p / \mathbb{F}_p)$
 $\ell(F/\mathbb{Q}_p) \cong \dim(\mathbb{F}_p / \mathbb{F}_p)$